

FLUX Mouse: A Seamless High-DOF Input Device for Intuitive 3D Interaction

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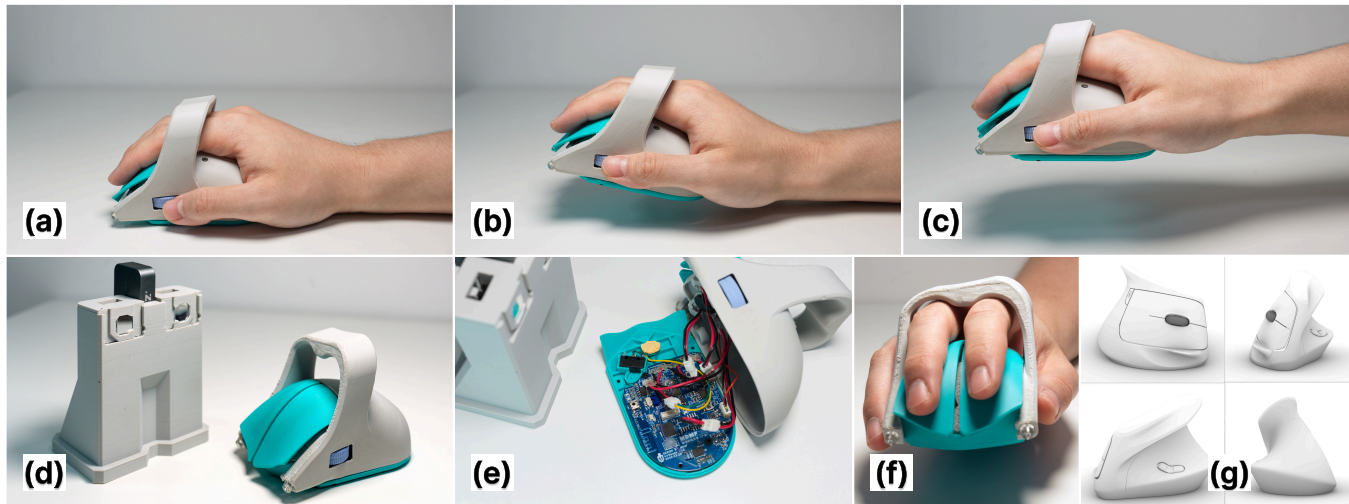


Figure 1: We introduce the FLUX Mouse, a hybrid input device with automatic mode transitions between traditional 2D and high-DOF 3D interaction. (a) When on a flat surface, it functions as a standard 2D mouse; (b) when slightly lifted, it enters a 3 DOF mode tracking linear motion; and (c) when fully lifted, it switches to full 6 DOF control for both translation and rotation in space. (d) The 3D modes rely on an integrated internal inertial measurement unit (IMU) and an external binocular infrared vision module for robust spatial tracking. (e) The custom PCB combines infrared LEDs, the IMU, and the 2D optical sensor. To support ergonomic in-air use, we designed multiple form factors, including (f) a finger arc and (g) an alternative support ledge design.

Abstract

Spatial controllers enable rich 3D interactions but often compromise the precision and ergonomic familiarity of traditional mice. Existing solutions typically remain specialized, forcing users to juggle multiple devices or adapt to unfamiliar form factors. We introduce the FLUX Mouse, a spatial input device that combines traditional mouse ergonomics with 3D interaction capabilities, featuring fully automatic transitions between 2D cursor control and 6 DOF spatial interaction. The FLUX Mouse integrates an optical sensor for 2D tasks with an inertial measurement unit and binocular infrared

vision tracking for 3D tracking, and uses adaptive pressure sensing to balance precision and speed. Preliminary user feedback suggests intuitive usability, a manageable learning curve, and positive acceptance, indicating the potential to support psychological needs as outlined by Self-Determination Theory across diverse applications from professional modeling to immersive gaming.

CCS Concepts

• **Human-centered computing** → **Pointing devices**.

Keywords

3D interaction, high-DOF input device, intuitive control, spatial input

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1 Introduction & Related Work

Spatial interaction with 3D virtual environments (VEs) has been actively researched since the emergence of 3D user interfaces, and has grown rapidly with the rise of commercial VR and AR technologies [2, 9]. A system’s degrees of freedom (DOF) describe its possible movements — 2 DOF in 2D (XY translation), and 6 DOF in 3D (translation and rotation along three axes) [19]. High-DOF input devices were designed to handle the additional DOF in 3D VEs [23, 25], spanning professional devices like the 3DConnexion SpaceMouse® for 3D modeling [1] to entertainment devices such as the Meta Quest VR controllers [15].

Studies suggest that 3D input devices generally outperform their 2D counterparts in spatial tasks, with increased efficiency, ease of use, and immersion [10, 22, 24]. However, user acceptance remains a critical factor, with research consistently finding higher acceptance for traditional mouse-like designs compared to more unconventional forms [3, 7, 11, 12, 24].

Beyond performance metrics, intrinsic motivation significantly influences interaction design [17]. Self-Determination Theory (SDT) suggests that fulfilling core psychological needs — competence, relatedness, and autonomy — drives user engagement, learning, and long-term use [16]. Studies indicate that intuitive, easy-to-use interaction methods can enhance users’ sense of competence and autonomy, leading to greater intrinsic motivation and engagement [4, 21]. This highlights the potential of well-designed 3D input devices to align with SDT principles.

Despite advancements, most existing high-DOF input devices are optimized or designed only for specific scenarios, such as professional 3D modeling and gaming [1, 13]. These devices are not fully integrated into 2D pointing scenarios, forcing users to frequently switch between their 2D mice and 3D devices, which leads to workflow fragmentation and reduced efficiency [14, 22]. Furthermore, devices such as the 3DConnexion SpaceMouse® [1] are generally unusable in everyday tasks. The lack of seamless integration between traditional 2D manipulation and 3D spatial control hinders broader adoption in hybrid creative and productivity workflows.

To address this gap, we introduce the FLUX Mouse, an ESP32-based input device supporting both 2 DOF and 6 DOF input modes within a traditional mouse form factor for better user acceptance. The FLUX Mouse combines an optical sensor for 2D cursor control with a fusion of an inertial measurement unit (IMU) and a binocular infrared (IR) camera system for full 6 DOF spatial tracking. It offers automatic, sensor-driven transitions between modes and adaptive pressure-sensing features, providing a versatile tool for blended 2D/3D tasks. By supporting seamless switching and reducing the need for multiple devices, the FLUX Mouse seeks to reduce workflow fragmentation, minimize context switching, and lower the barrier to adoption for spatial interaction in hybrid creative and productivity tasks.

2 FLUX Mouse: Design and Implementation

Based on the analysis of prior work and existing solutions, we established the following key design principles:

- **Seamless Mode Transitions** The device should support easy and intuitive transitions between traditional 2D and

high-DOF spatial input modes, enabling fluid workflows without explicit user intervention.

- **Universal Applications** The device should support diverse scenarios from professional software (e.g., Blender) to entertainment (e.g., gaming) and other common daily tasks, reducing the need to switch devices.
- **Intuitive Control** The device should minimize learning curves, leveraging familiar mouse ergonomics and natural input mappings.

2.1 Hardware and Sensing

2.1.1 External Layout. The FLUX Mouse is a custom 3D input device designed to enable embodied and intuitive interaction. The device maintains a traditional mouse form factor, featuring two separate main buttons, a side scroll wheel, a main switch at the bottom, and a USB-C port for charging and wired connection on the right side. Two IR LEDs are placed at the front edge for visibility by the external tracking system (see Figure 2). As shown in Figure 1(f), a specially designed finger arc covers the user’s index and middle fingers to enhance grip stability and reduce fatigue during in-air use.

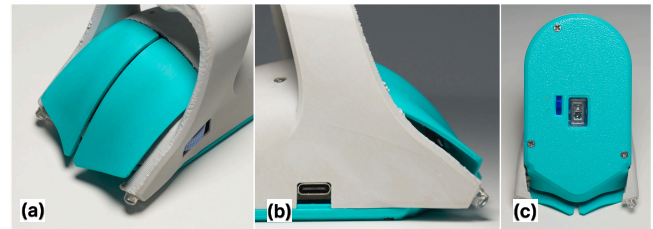


Figure 2: External layout of the FLUX Mouse, (a). main buttons, side scroll wheel, and front IR LEDs; (b). USB-C port on the right side; (c). main switch at the bottom.

2.1.2 Internal Components and Circuit Integration. The FLUX Mouse uses an ESP32-S3R8 microcontroller (MCU) for onboard processing, Bluetooth® 5 Low Energy (BLE) wireless communication, and USB connectivity, with GPIO pins supporting SPI digital interfaces for sensor data transmission [6]. The PAW3805EK optical sensor provides high-resolution, all-surface detection for 2D input mode [8]. For high-DOF interactions, a BNO085 IMU [5] captures linear motion, while an external ESP32-driven binocular IR vision system [20] tracks rotational movements via two IR LEDs at the front. A pressure sensor is placed under the main buttons to detect the user’s pressing force during operation. The arrangement of PCB and all major internal components is shown in Figure 3.

2.2 Data Processing and Software Implementation

To achieve stable high-DOF tracking, we implemented a multi-sensor hardware solution with a modular software stack, as shown in Figure 4. The system integrates two calibrated binocular vision sensors for vision-based tracking (20-30 Hz), along with an onboard IMU and optical sensor for high-frequency inertial (100 Hz) and surface motion tracking. These data streams are then processed in

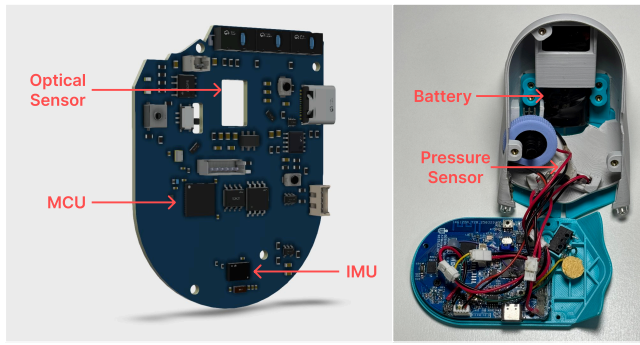


Figure 3: PCB and internal layout of the FLUX Mouse

a fusion module, with complementary filtering to combine visual and IMU information, while a sliding window average reduces jitter and high-frequency noise. The fused pose data are stored in shared memory and synchronized with subscribed applications (e.g., Blender, Unity 3D) using Windows event signaling. A PyQt-based graphical user interface allows users to monitor device status, track performance, and adjust parameters in real-time, improving both usability and debugging.

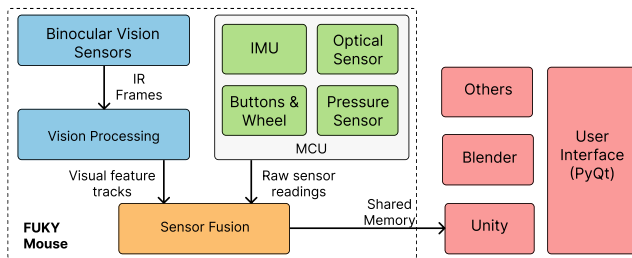


Figure 4: System Architecture of the FLUX Mouse

2.3 Ergonomics

The FLUX Mouse replicates traditional mouse ergonomics to reduce grip force and user fatigue during in-air use, enhancing intuitive control.

Finger Arc The current product features a finger arc that distributes the weight evenly across the index and middle fingers, maintaining the center of mass near the palm during in-air interactions, enabling stable control with minimal lateral finger pressure for button activation and reducing the possibility of slippage during repositioning.

Support Ledges In the next prototype iteration, we plan to implement support ledges — thin extensions on the mouse’s sides and top designed to enhance ergonomic control. These ledges naturally rest against the thumb-index webbing and little finger, aligning with the wrist’s biomechanics to allow $\pm 34^\circ$ horizontal tilting without requiring wrist rotation. This configuration is ideal for high-mobility scenarios such as gaming, where extended, dynamic movement is common.

2.4 Input Modes

The FLUX Mouse supports 2 DOF, 3 DOF, and 6 DOF input modes, with automatic transitions triggered by motion detected via the IMU.

2 DOF Mode Activated when the optical sensor detects surface contact. In this mode, only movement data collected by the optical sensor is transmitted to control the cursor. The default resolution of this mode is 1500 counts per inch (CPI), configurable up to 3000 CPI. This mode offers high user familiarity and minimal fatigue, functioning like a traditional mouse.

3 DOF Mode This is an intermediate state between 2 DOF and 6 DOF modes, which is active when the hand is lifted slightly while the wrist maintains contact with the surface. In this mode, linear motion along three axes captured by the IMU is used to control movement in the environment. This mode provides a more intuitive interaction in 3D VEs, as the user’s hand movements directly translate to linear movements within the scene.

6 DOF Mode This mode is activated when both the hand and wrist are lifted from the surface, enabling full spatial control similar to a VR controller, and is ideal for actions such as rapid perspective changes and 3D model rotation.

2.5 Interaction Design & Key Features

The FLUX Mouse enhances 3D interaction in virtual environments (VEs) through several novel design choices:

Pressure-Sensitive Dynamic Control: A pressure sensor is placed under the buttons that allows users to dynamically adjust the sensitivity of the operations. Applying more force increases the manipulation sensitivity for rapid, large-scale adjustments of the 3D objects, while reduced pressure enables finer, precise control. This principle applies across translation, scaling, and viewport navigation in the 3D space.

Translation and Scaling: Physical movement of the mouse directly maps to object displacement in the 3D space for translation. For scaling, physical mouse movement determines the object’s scaling amount, and the direction of the movement controls the orientation of the scaling. Both operations leverage pressure sensitivity for adjustable feedback.

Rotation: The FLUX Mouse allows users to rotate virtual objects without needing to perform corresponding physical rotations of their own hand. Instead, users manipulate a "Point of Interest" (POI) in 3D space using translational hand movements, and the virtual object automatically orients itself in relation to this POI. This decouples the need for physical hand rotation from object rotation, allowing users to control the feedback scale by adjusting the POI’s position, enabling both broad gestures and fine-tuned adjustments with varying degrees of hand movement.

Combined Transformation Operations: The FLUX Mouse supports simultaneous control of multiple transformations (e.g., rotation and scaling, or translation and rotation) using its 3 DOF input. This streamlines complex manipulations that would typically require sequential operations.

Viewport Navigation: Camera control is simplified in viewport navigation with the FLUX Mouse: users can easily reposition the camera’s POI in the 3D space, leveraging the higher DOF. Simultaneously, they can orbit around this POI and adjust the zoom level

in a single, compound action with pressure sensitivity for sensitivity control. This reduces the steps needed for common viewport adjustments compared to traditional 2D methods.

3 Evaluation

To obtain early-stage feedback on usability and perceived intuitiveness, we conducted a formative, qualitative evaluation with a beta tester (male, 23), with several years of gaming and 3D modeling experience, but no prior exposure to high-DOF devices of this form.

3.1 Procedure

The participant was first given a brief tutorial on the device's functions and key features, and then interacted with the FLUX Mouse prototype in a game-styled demo Unity scene featuring common 3D manipulation tasks: selection, translation, and rotation. After completing the tasks, the participant took part in a semi-structured interview focusing on intuitive control based on two questions from the Player Experience of Needs Satisfaction (PENS) [18]:

- Q1: Learning the game controls was easy.
- Q2: The game controls are intuitive.

3.2 Findings

3.2.1 Ease of Control Learning. The participant reported that "brief guidance and a practice scene are needed to fully understand the device's purpose." After a brief tutorial, the participant quickly performed the basic set of actions, suggesting the overall learning curve is moderate and dependent on in-game text-based guidance.

3.2.2 Intuitive Control. The participant described the interaction as "straightforward, similar to a VR controller, emphasizing what-you-see-is-what-you-get." The sensor fusion 6 DOF solution and the interaction design were immediately comprehensible, and the mouse form factor was reported to be intuitive. The automatic input mode transition was also reported as "better than expected." During manipulation, the user's attention naturally stayed on the virtual object rather than on the mechanics of switching, which the participant described as "did not feel awkward at all."

4 Discussion & Future Work

The formative evaluation suggests that the FLUX Mouse is capable of unifying 2 DOF to 6 DOF interaction intuitively using a familiar mouse form factor, providing seamless mode transitions and intuitive spatial control. The beta tester's positive feedback aligns with SDT principles [16], suggesting that the mouse-like ergonomics and pressure-adaptive scaling may enhance perceived competence through reduced fatigue and intuitive spatial mapping.

However, several limitations persist in our current prototype. The current sensor fusion solution introduces some latency, especially during rapid movements, and the single-participant, qualitative evaluation significantly limits the generalizability of our study.

Future work will focus on the following directions:

- **Algorithm Optimization** Implement advanced on-device filtering algorithm (e.g., Kalman filtering) to reduce noise, lower latency, and enhance tracking performance and robustness.

- **Ergonomic Iteration** Implement the support ledge design to further improve comfort and to test and compare with the current design for improved ergonomics.
- **Further Evaluation** Conduct a comprehensive, mixed-methods user study with more participants (approximately 30) in both professional software (Blender) and entertainment scenarios (custom Unity game levels), and incorporating standardized usability and workload measures.

5 Conclusion

We presented the FLUX Mouse, a high-DOF input device that combines traditional mouse ergonomics with seamless, intuitive spatial interaction. A formative evaluation provided early indications of usability and intuitive control, but also highlighted limitations and areas for improvement. Our findings, while preliminary, suggest that sensor-driven mode switching and pressure-adaptive control hold promise for hybrid 2D/3D workflows. We will address the identified limitations and conduct broader studies in future work to further validate and refine the FLUX Mouse design.

Acknowledgments

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